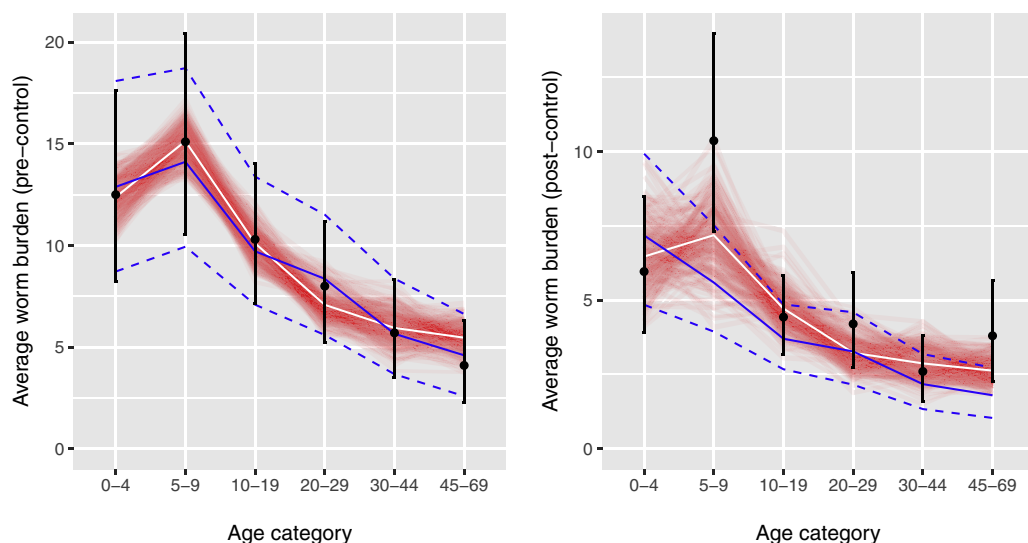
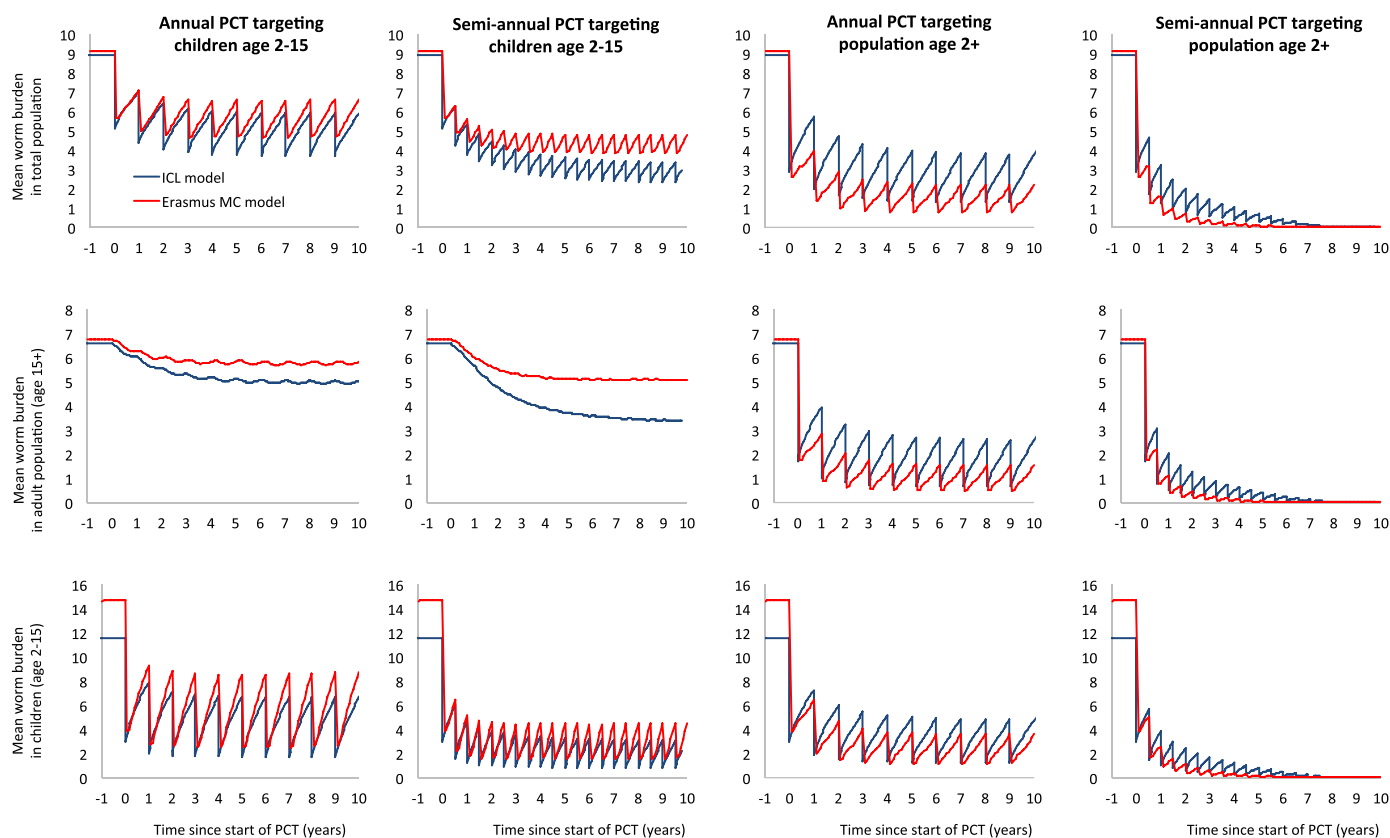




**Fig. 1.** Comparison of model predictions for *Ascaris lumbricoides* infection to data. The deterministic ICL model (blue) and the stochastic individual-based Erasmus MC model (white) were calibrated to reproduce data on age patterns in pre-control infection levels (black bullets with 95% confidence intervals, left panel) in Pulicat, India (Elkins et al., 1988, 1986). Next, forward predictions were made given information on the timing and coverage of mass drug administration. The right panel compares the reported worm expulsion data and the predicted age profile in infection levels at 11 months after a single round of mass drug administration with pyrantel pamoate targeting all individuals between age 1 and 70, implemented at 68% coverage of the overall population, assuming that the drug kills 95% of the worms in treated individuals (Elkins et al., 1988, 1986). Predictive uncertainty (i.e. model parameter uncertainty) for the ICL model is represented by the dashed blue lines, which represent 95%-Bayesian credible intervals; for the Erasmus MC model, predictive uncertainty (i.e. stochastic and model parameter uncertainty) is represented by a set of 250 stochastic simulations (red lines) that approximate the predictive posterior distribution (see Methods section for details).



**Fig. 2.** Comparison of model predictions for the impact of different preventive chemotherapy strategies on the *A. lumbricoides* worm burden in different age groups. Both models were calibrated to predict a pre-control average female worm load of 9 worms per person in the general population, conditional on the assumption that the level of parasite aggregation is  $k_w = 0.8$ . Further, in both models seventy-five percent of the PCT target population were assumed to be treated with albendazole, which was assumed to kill 99% of the worms in treated people. The parameters for worm biology and age patterns in exposure and contribution to the reservoir are as described in Supplemental Table S1 and differ between the two models.